

Lesson 3 ROS2 Command-Line Operation

1. Component of ROS2 File System

The ROS2 file consists of Packages and Manifests.

Packages are the basic organizational unit in ROS2 software, consisting of a collection of related files and directories used to organize and manage ROS2 nodes, libraries, and resources. These packages contain source code for nodes, configuration files, message and service definition.

Manifests (package.xml) is a description file for the packages used to define the relevant metadata and dependencies of the package. In ROS2, packages typically contain a file named package.xml, which contains metadata information about the package, such as name, version, maintainer, license agreement, and dependencies.

Manifest file (package.xml) plays a crucial role in ROS2, providing descriptions and management for packages. This enables the ROS2 to properly handle dependencies between packages and perform appropriate building and performance.

2. Get to Know ROS2 Basic Terms

The following table will introduce you some basic terms in ROS2:

Term	Instruction
Underlying communication layer (DDS)	It provides high-performance, reliable, real-time data communication and integration capabilities, thereby establishing fundamental support for messaging and service invocation between nodes in ROS2.
Node	Node is the smallest unit of processing running in ROS. It is typically an executable file. Each node can use topics or services to communicate with other nodes.
Message	The variables of data types such as int, float and boolean.
Topic	A one-way asynchronous communication mechanism. By publishing messages to topics or subscribing to topics, the data transmission between nodes can be realized. The topic type is determined by the type of corresponding message.
Publishing	Send data with a message type corresponding to the topic content.
Publishers	For publishing to take place, the publisher node registers various information such as its topics on the master node, and transmits messages to subscribing nodes that wish to subscribe.

Subscribing	Receive data with a message type corresponding topic content.
Subscribers	For subscription to take place, the subscribing node registers various information such as its topics on the master node. Subsequently, it receives all messages from publisher nodes that have published topics of interest to this node, via the master node.
Services	A bidirectional synchronous communication mechanism where the service is provided to the client request corresponding to a specific task and service servers gives the service response.
Service Servers	A node taking requests as input, and providing responses as output.
Service Clients	A node taking responding as input, and providing requests as output.

3. Get to Know ROS2 Commonly File

The following table will provide explanations for the commonly used files in ROS:

Term	Instruction
Urdf file	A model file describing robot's entire elements, including link, joint, kinematics parameters, dynamics parameters, visual models and collision detection models.
Srv file	It is stored in the srv folder used to define ROS service messages, consisting of two parts: request and respond. The request and respond are separated by the "---" symbol.
Msg file	It is stored in the msg folder used to define ROS topic messages.
package.xml	Description of the package attributes, including the package name, version number, authorship and other information.
CmakeLists.txt	Compile the configuration file using Cmake.
launch	Launch files contain system-wide instructions for launching nodes and services required for the robot to operate.

4. Get to Know Commonly Used Command

4.1 Package Management Tool

The ros2 pkg has a total of five commands: create, executables, list, prefix, xml.

Command	Instruction
ros2 pkg create	Create packages, specify package name, build type, dependencies, node names, etc.
ros2 pkg list	View the list of packages in the system.
ros2 pkg executables	View a list of executable files within packages in ROS2.
ros2 pkg prefix	View the the installation path prefix of a specific package in ROS2.
ros2 pkg xml	View the contents of the .xml file for the specific package.

4.2 Node Running Tool

The function of ros2 run is allow you to run nodes within ROS2 package. The basic syntax is “**ros2 run <package_name> <node_name>**” as pictured.

```
ubuntu@raspberrypi:~$ ros2 run turtlesim turtlesim_node
StandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-ubuntu'
[INFO] [1705999223.034178002] [turtlesim]: Starting turtlesim with node name /t
urtlesim
[INFO] [1705999223.044200929] [turtlesim]: Spawning turtle [turtle1] at x=[5.54
4445], y=[5.544445], theta=[0.000000]
libGL error: glx: failed to create dri3 screen
libGL error: failed to load driver: vc4
```

4.3 Command-line Tool

The ro2 node command is used to view node messages.

Command	Instruction
ros2 node list	View the list of active nodes within the current domain (nodes grouped by the same ROS_DOMAIN_ID)
ros2 node info	View the detailed information about a specific node in ROS2, including its subscriptions, publications, publishing, services and actions, etc.

4.4 Topic Operation Tool

The ros2 topic command in ROS2 is used to perform operations related to topics.

Command	Instruction
ros2 topic list	List all active topics within the domain.
ros2 topic info	Provide information about a specific topic, including its type, and publishers/subscribers.

ros2 topic type	View the type of topic messages.
ros2 topic find	Find the topics related to a specific message type in ROS2.
ros2 topic hz	Display the average publishing frequency of a topic in ROS2.
ros2 topic bw	Calculate the bandwidth usage of a specific topic.
ros2 topic delay	Calculate the message delay using the timestamp in the message header in ROS2.
ros2 topic echo	Display the topic message in console.
ros2 topic pub	Publish a message to a specific topic through command-line.

4.5 Interface Operation Tool

The ros2 interface command is used to perform operations related to the interfaces in ROS2.

Command	Instruction
ros2 interface list	List all available interfaces in the system, including Messages, Services, and Actions.
ros2 interface package	List the child interfaces within the specific package.

ros2 interface packages	List packages containing interfaces.
ros2 interface show	Display the detailed contents of the specified interface.
ros2 interface proto	Display message module.